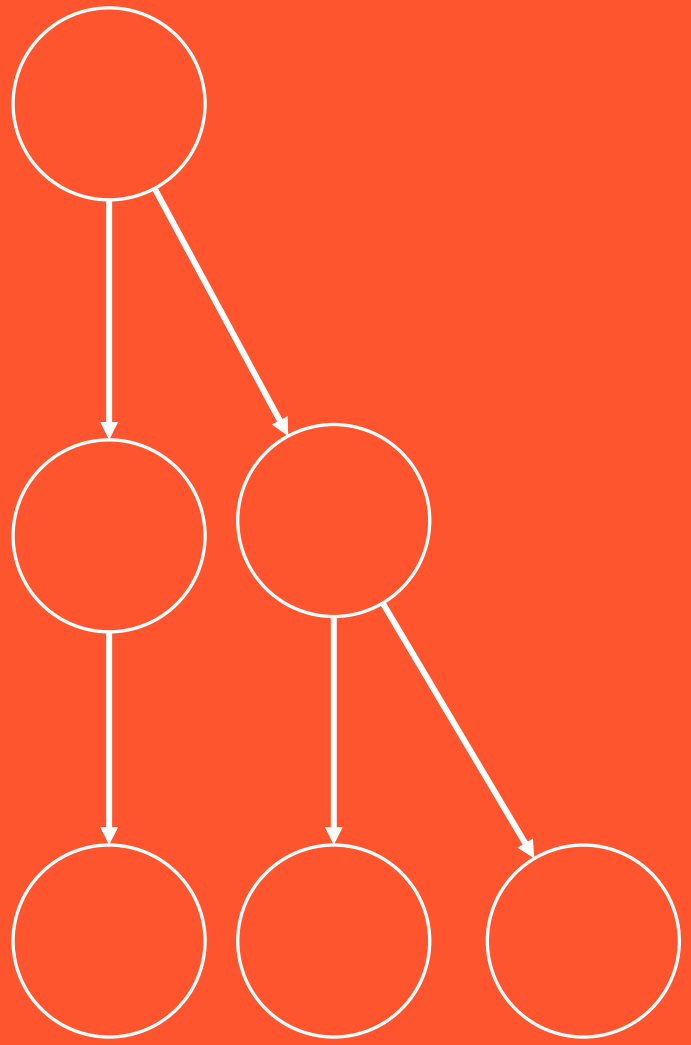


BCOG 100

Perception and Action

Lecture 8.3 — Receptor Sheets, Maps, and the Cerebellum

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

To Start: What Would It Take?

In your group, three minutes.

Every procedure in the second lecture could run in any substrate. Pick one: the delay line, a copy of the command, or learning a mapping from mismatch.

What would a nervous system need, to run it? What kind of cells, wired how, and how fast?

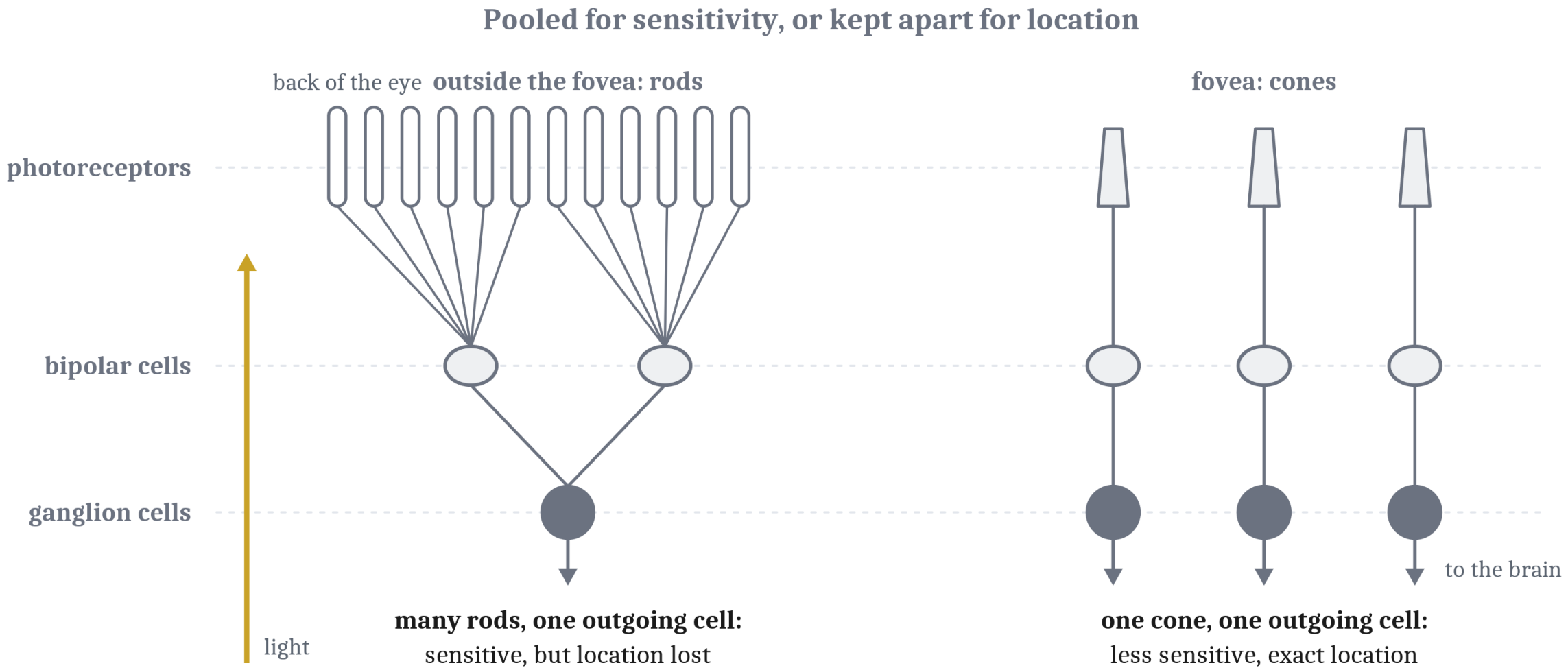
From Receptor Sheet to Cortex

Most receptor sheets keep their layout as maps: **retinotopy**, **tonotopy**, **somatotopy**. Of the senses that reach the cortex, all but smell pass through the thalamus.

Sense	Receptors laid out by	What the layout gives free	Route to the cortex
Vision	Direction in the visual field	Direction, relative to the eye	Thalamus (lateral geniculate nucleus), then primary visual cortex
Hearing	Frequency	Pitch, not direction	Brainstem stations, thalamus, auditory cortex
Balance	Axis of rotation, direction of tilt	Which way the head turns or tilts	Brainstem and cerebellum; thalamus to several areas, no single primary area
Touch and proprioception	Location on the body	Where on the body	Spinal cord and brainstem, thalamus, somatosensory cortex
Smell	Scattered, then sorted by receptor type	Nothing about location	Olfactory bulb straight to olfactory cortex
Taste	Across the tongue, every region sensing every taste	Which of five basic tastes	Brainstem, thalamus, taste cortex
Lateral line and electric sense (fishes)	Along the body surface	Where along the body	Cerebellum-like structures in the hindbrain, then the midbrain

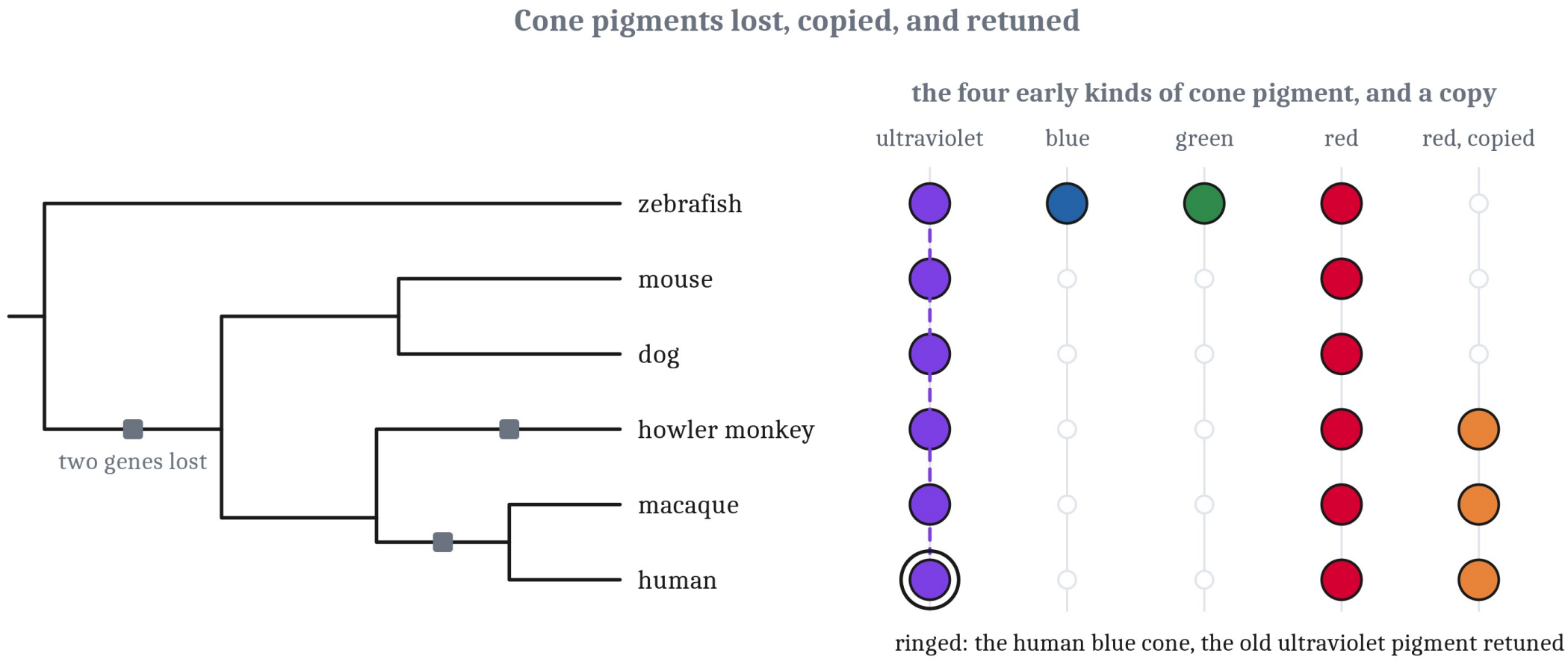
The Retina Computes

A **photoreceptor** is the retina's receptor: a **rod** for dim light, a **cone** for daylight, detail and color. Wiring trades sensitivity for location.



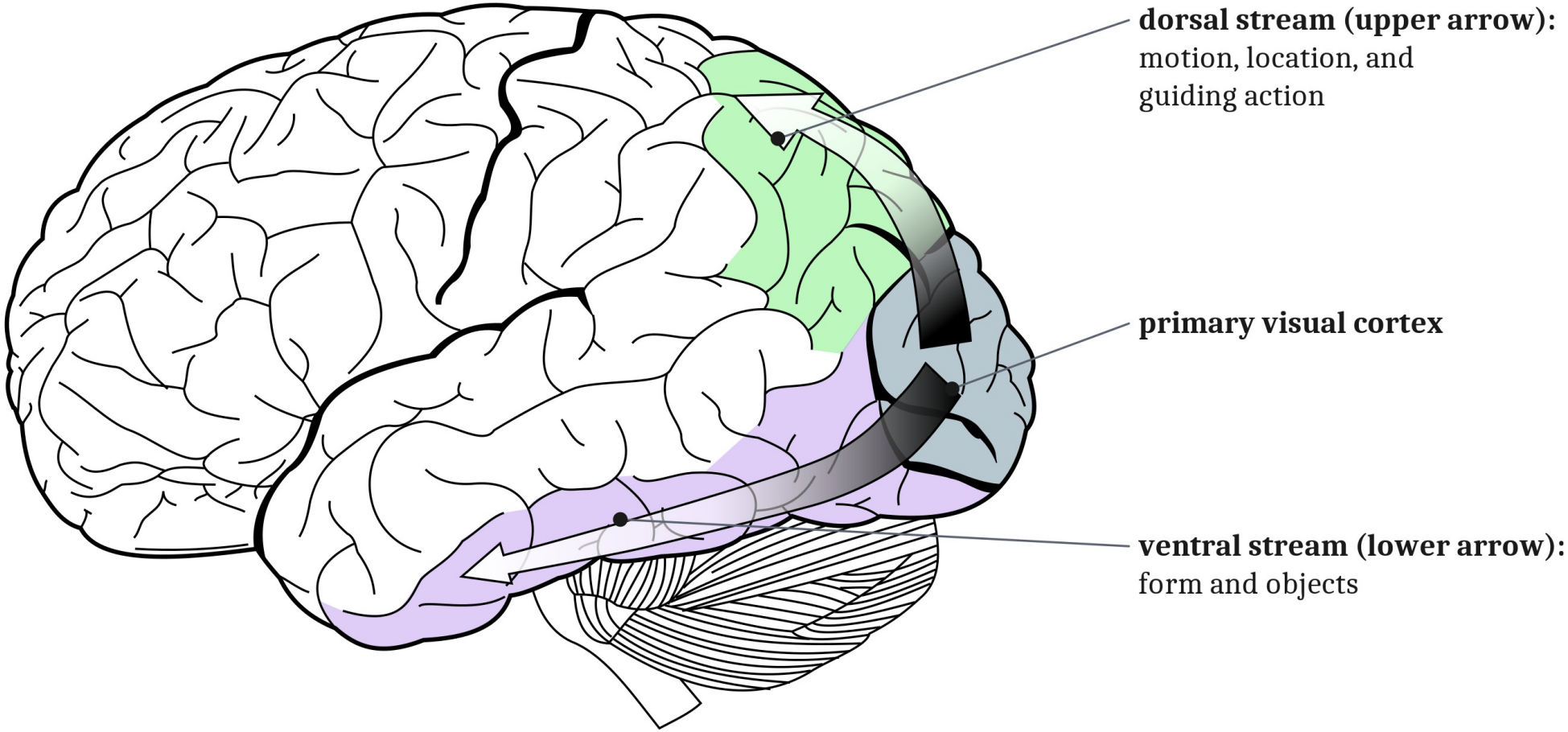
Cone Classes, Lost and Duplicated

Early vertebrates had four cone classes; mammals lost two. A third cone in our ancestors came from a duplicated gene, and the human blue cone carries the old ultraviolet pigment.



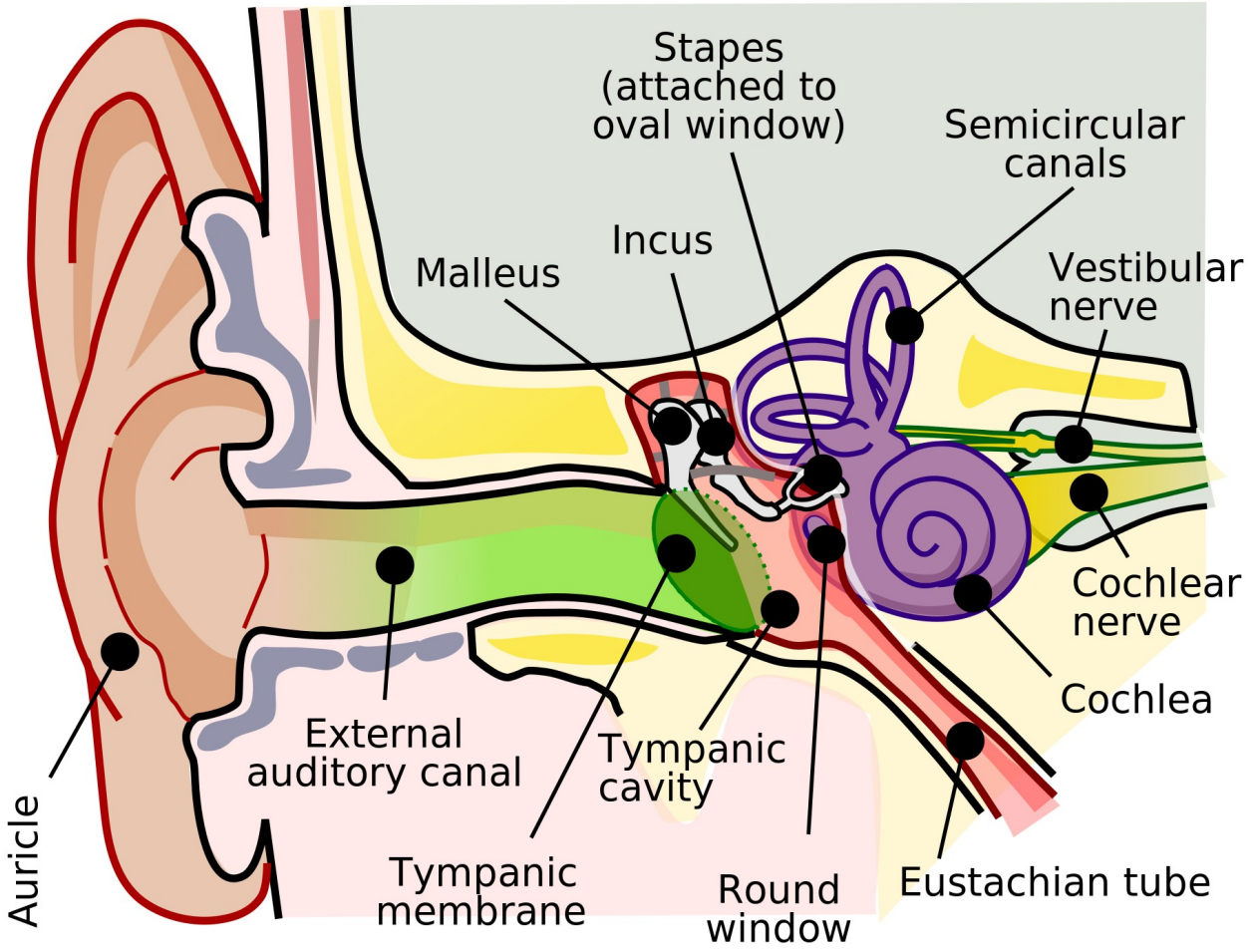
Two Streams, Specialized

The ventral stream for form and objects; the **dorsal stream** for motion, location and guiding action. Specialized, not independent.



The Ear, from Old Parts

The middle ear's bones began as a jaw brace and a jaw joint. An eardrum concentrates the force of sound in air, and eardrums evolved late, separately, at least five times.



Its labels, in this chapter's words

Auricle
outer ear

Tympanic membrane
eardrum

Tympanic cavity
middle ear

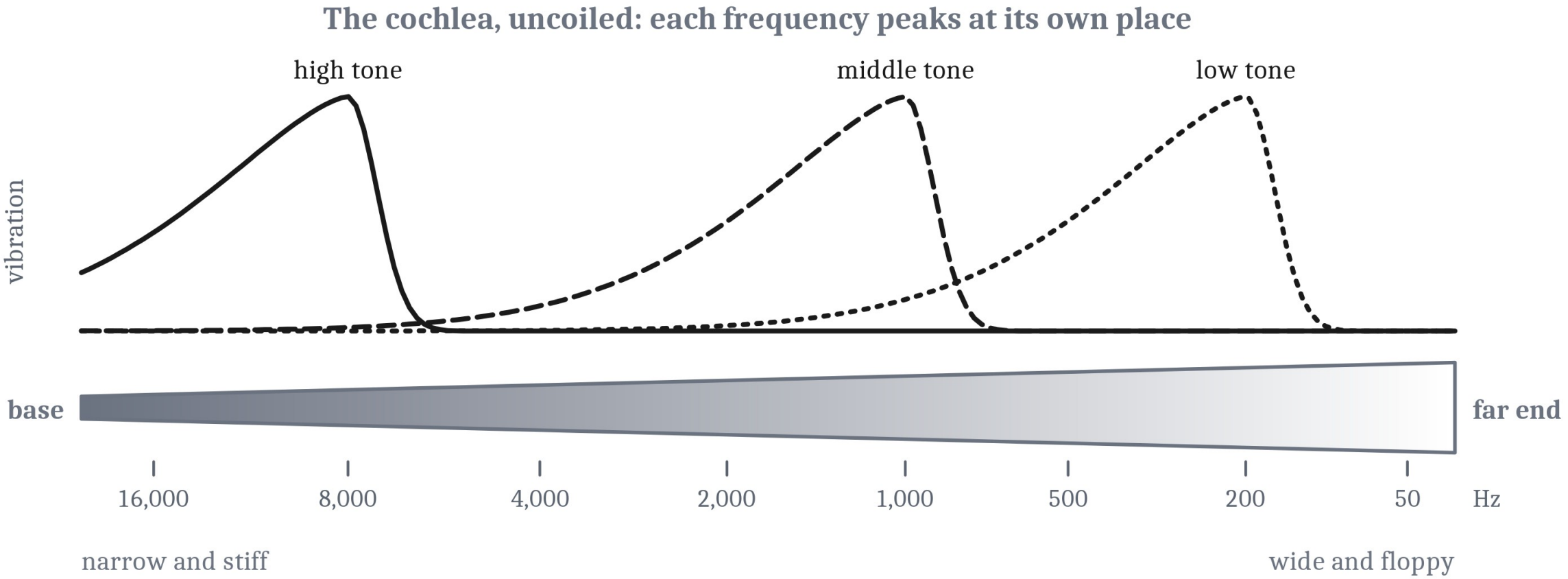
Malleus, incus, stapes
the three bones

Eustachian tube
tube to the throat

Semicircular canals
balance organs

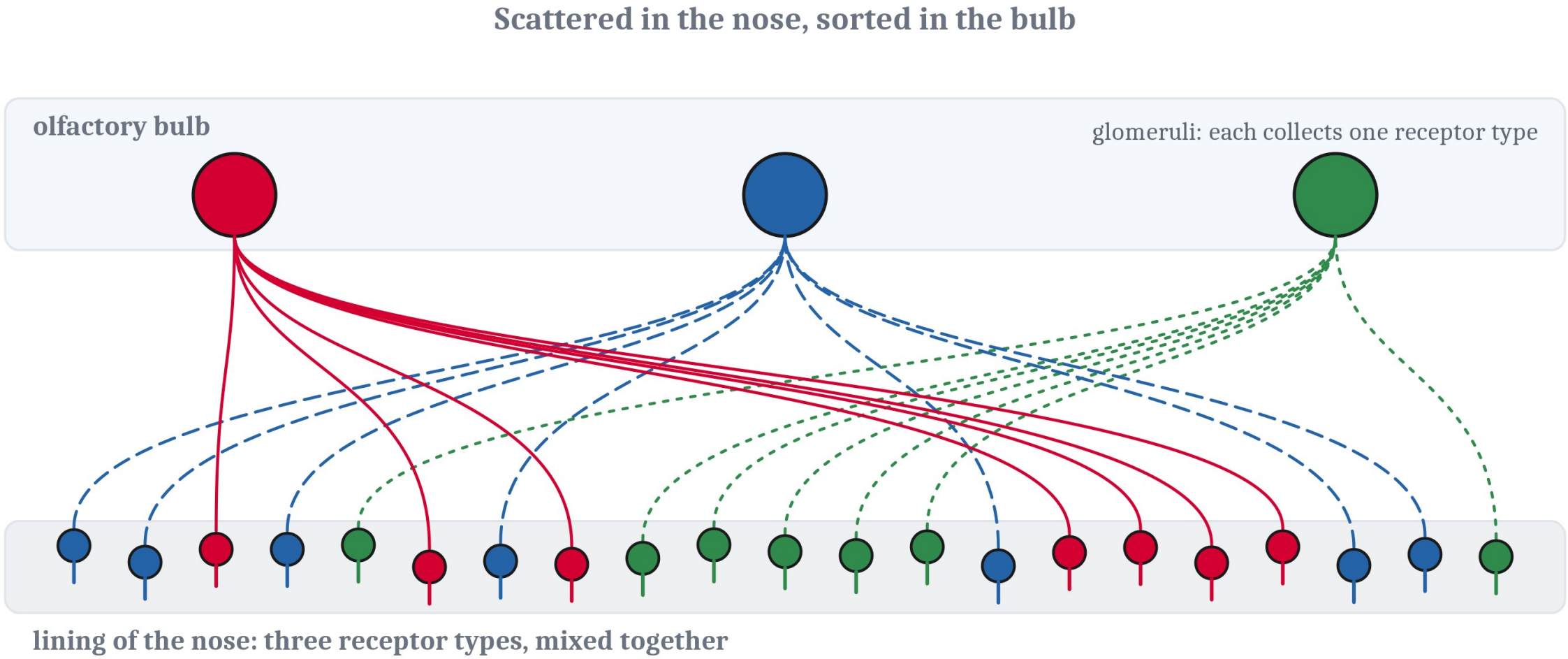
The Cochlea, the Hair Cell, and the Owl

The cochlea is a bank of filters by pitch. The **hair cell** opens its channels directly, fast enough for timing. The **superior colliculus** aligns auditory space with visual space.



Smell and Touch

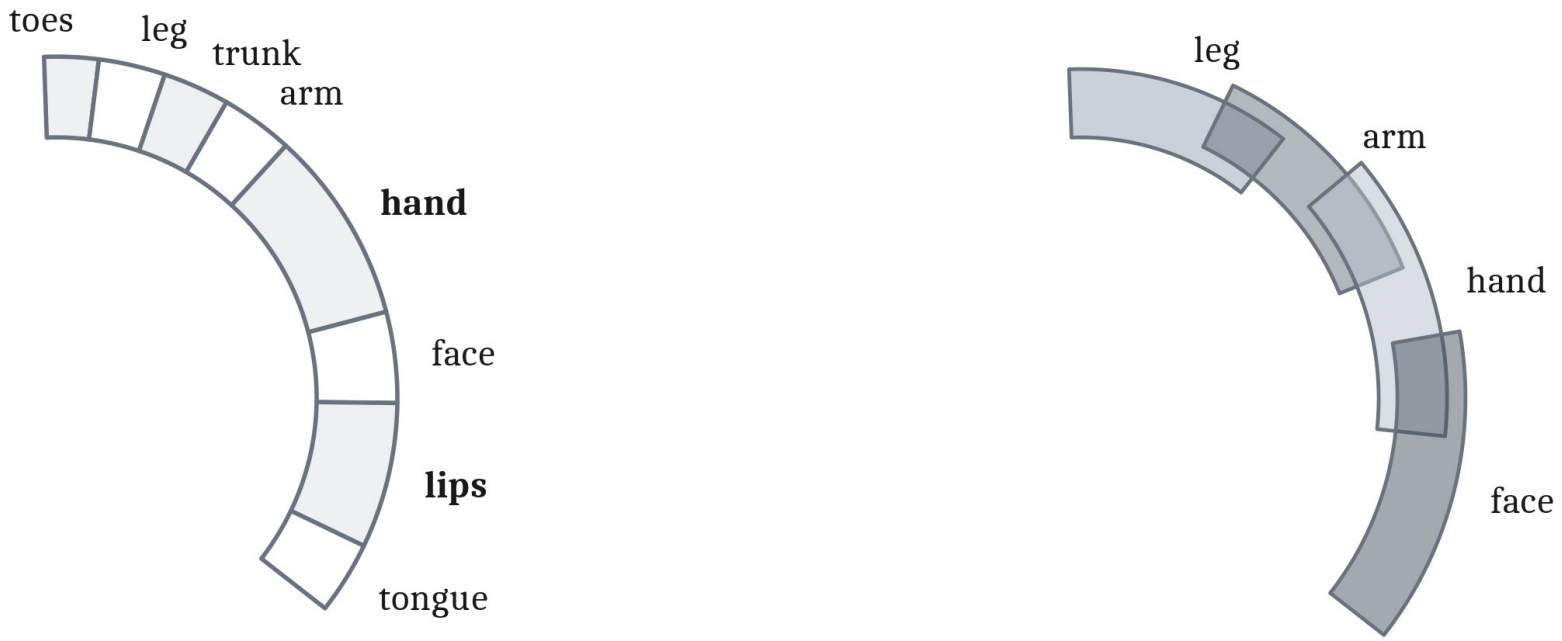
Smell neurons of one receptor type, scattered across the nose, converge on the same few glomeruli. A **glomerulus** is a small cluster of connections in the olfactory bulb. Touch keeps location, mapped as a **homunculus**.



Motor Cortex Is Not a Mirror

Motor cortex maps the body coarsely, in overlapping zones organized around what the body does. Direction is a vote across many neurons.

Two maps of the body, side by side in the cortex



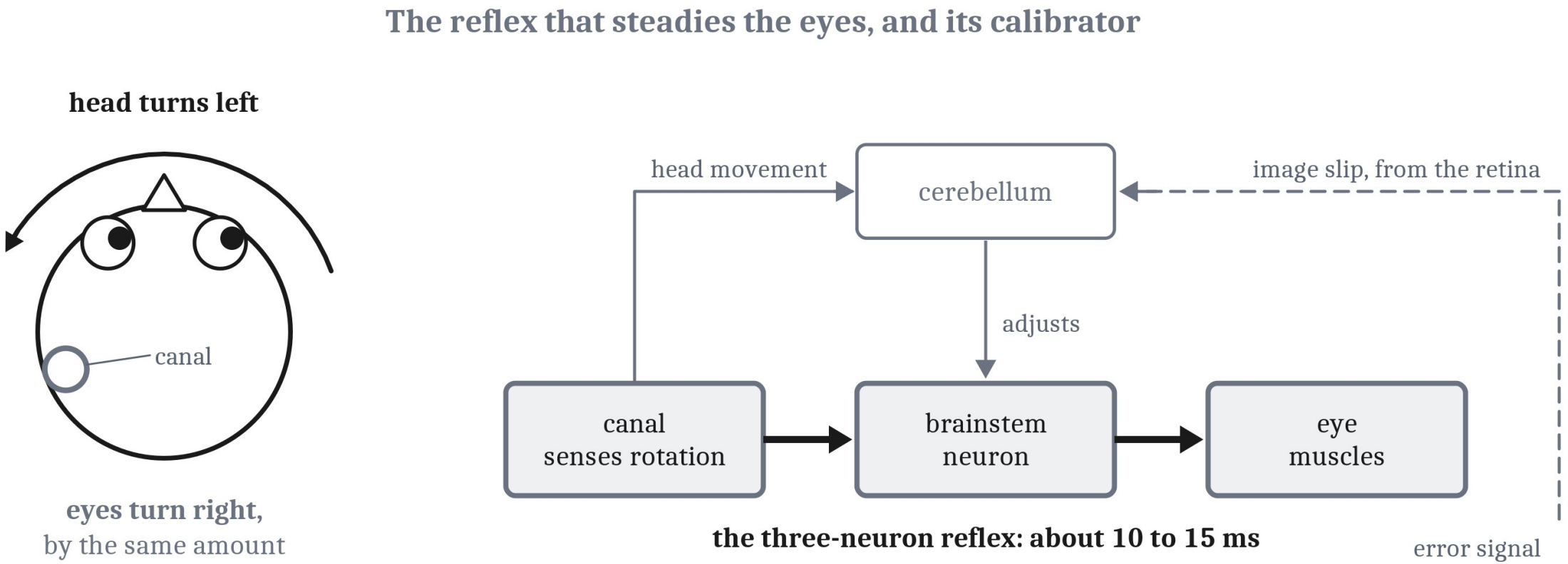
somatosensory map
each part in its own segment

motor map
the same order, in overlapping zones

each strip is a slice of cortex, running from the top of the head down the side

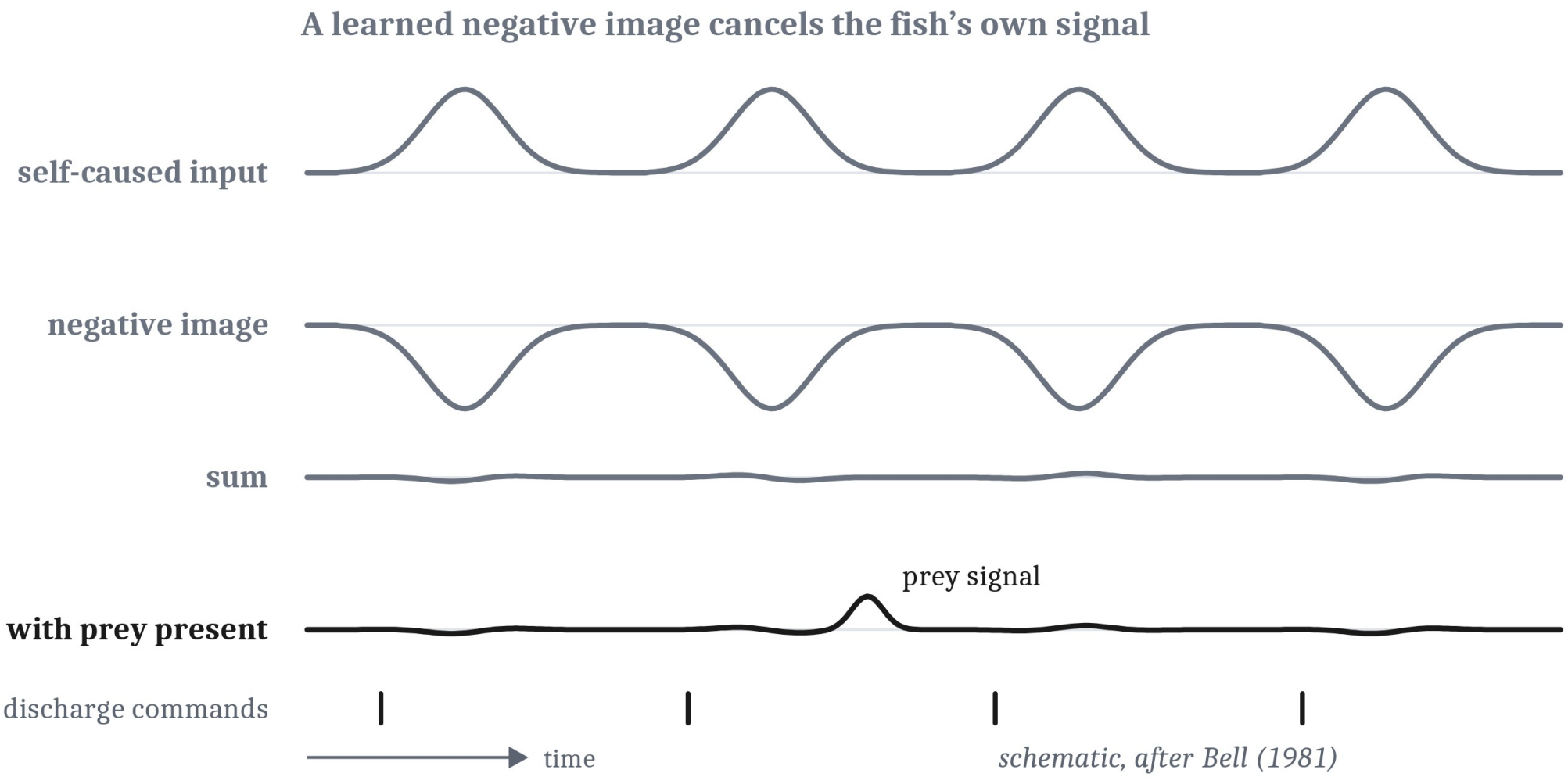
The Cerebellum Calibrates

The **vestibulo-ocular reflex** keeps the eyes steady as the head turns, without waiting for vision. Recalibrating it from the image's slip needs the cerebellum.



The Electric Fish’s Negative Image

A **negative image**, opposite in shape to the expected self-caused input and driven by the command, cancels it and leaves the prey. The rule is Hebb’s, reversed.



What Every Sense Does

Perception is not a recording. It manages the ill-posed problem well enough to act on.

What things are

Across the variation each sense's own physics produces.

Met with tolerance, not perfect invariance.

The object from a new angle; the ringtone in a pocket.

Where things are

From a receptor sheet's layout, or by comparing two sensors: two ears for direction, two eyes for distance.

A place on a sheet means nothing until combined with where the sheet pointed.

World or self

Dim the sense, measure the movement, or predict and subtract.

The prediction also beats delay.

Learned, continually, from mismatch.

Perception settles for what its signals constrain well, adds assumptions that are usually true,

and corrects itself from its errors: managed well enough to act on, not solved.

Reading Motor Cortex

In your group, four minutes.

Brain–computer interfaces record from motor cortex to let a person with paralysis move a cursor or a robotic arm.

Why record from many neurons rather than find the one for each direction? And why might the system need recalibrating, day after day?

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